

Bhuvaneshwari Mallichetty Chandrasekaran

Semiconductor & Quantum Device Researcher - Nanofabrication · Characterization · Si/Ge Quantum Dots

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PROFILE

Passionate about both semiconductor and quantum device research, with hands-on experience in cleanroom fabrication, device characterization, and simulation. Seeking a PhD or R&D opportunity in Europe to work across the full device development workflow.

EDUCATION

MSc Nanotechnology, Electrical Engineering and Computer Science <i>KTH Royal Institute of Technology, Stockholm, Sweden</i>	Aug 2021 – April 2026
B.E., Electronics and Communication Engineering <i>Anna University, India</i>	Aug 2016 – May 2020

TECHNICAL SKILLS

Nanofabrication	PECVD, ALD, Photolithography, RIE (SOI CMOS process flow, NFET/PFET integration)
Characterization	Micro-PL spectroscopy, SEM, AFM, I-V/C-V electrical characterization, SPICE compact model fitting
Simulation	TCAD (PADRE/Prophet), COMSOL, SOI & bulk MOSFET modelling
Computational	Python, Qiskit, Qiskit Metal, QuTiP - qubit decoherence, Rabi oscillations, Josephson junction modelling
Instrumentation	LabVIEW-based data acquisition, automated test routines, sensor calibration

RESEARCH EXPERIENCE

Master Thesis Student - Optical Characterization of Si Quantum Dots <i>KTH Royal Institute of Technology</i>	Feb 2023 – Feb 2024
- Characterized luminescence behaviour of individual Si quantum dots on SOI substrates using single-dot micro-PL spectroscopy, resolving emission spectra across individual dots (630–676 nm, FWHM 24–39 nm). - Quantified the effect of High Pressure Water Vapour Annealing (HWA) across three pressure conditions (1.3, 2.6, 3.9 MPa), measuring up to 12% improvement in QD duty cycle and a 0.04 Hz reduction in on/off blinking frequency, linking oxide-layer engineering directly to quantum dot stability. - Connected surface passivation physics - dangling bond suppression and non-radiative recombination reduction - to coherence mechanisms in semiconductor spin qubits, bridging optical characterization to quantum device engineering.	
Student - Characterization Lab <i>KTH Royal Institute of Technology</i>	Aug 2021 – Dec 2022
- Performed SEM and AFM analysis on nanofabricated structures, reducing surface roughness measurement uncertainty from 41 nm to 14.74 nm through systematic scan optimization and parameter tuning. - Conducted I-V and C-V electrical characterization of semiconductor devices, extracting ideality factor ($n = 3.90$), series resistance (6.7Ω), and doping concentration ($4.69 \times 10^{13} \text{ cm}^{-2}$) using SPICE-based compact model fitting. - Applied photoluminescence spectroscopy to assess optical device performance, producing quantitative reports connecting nanoscale material properties to device behaviour.	
Student - Nanofabrication Lab <i>KTH Royal Institute of Technology, Stockholm County, Sweden</i>	Aug 2021 – Dec 2022

- Executed multi-step SOI CMOS process flow in an ISO-class cleanroom, performing PECVD, ALD, photolithography, and RIE to fabricate **0.96 μm^2 contact hole** geometries with **12 nm dielectric films**, minimizing parasitic capacitance for high-quality device integration.
- Performed wafer inspection, film characterization, and process validation across NFET and PFET device integration steps, maintaining process repeatability and documenting results for continuous improvement.
- Fabricated and characterized a MEMS-based thin-film aluminum RTD temperature sensor (photolithography, oxidation, sputtering, lift-off, DRIE); validated performance via Wheatstone bridge measurement, achieving experimental TCR close to the theoretical value for aluminum.

Intern, Test Facilities & Electronics and Instrumentation System Group
Indian Space Research Organisation (ISRO), Satish Dhawan Space Centre

May 2019 – Jun 2019

- Improved sensor calibration accuracy by **20%** through development and validation of LabVIEW-based data acquisition workflows and automated test routines for research instrumentation.
- Supported static testing of solid rocket motors through instrumentation setup, sensor calibration, data recording, and signal processing during live test campaigns.

Independent Research Projects (self-initiated)

- AlGaIn/GaN polarization and 2DEG modelling in Python.
- Quantum device modelling: T1/T2 decoherence, Rabi oscillations, Josephson junctions, quantum dot blinking statistics (Qiskit, Qiskit Metal, QuTiP).
- Semiconductor device structure simulation using Python, bridging simulation with experimental device physics.

SIMULATION & DEVICE MODELLING PROJECTS

- **Nano-CMOS MOSFET Scaling Study** - Simulated threshold voltage roll-off, DIBL, sub-threshold swing, and leakage current across bulk and SOI MOSFETs under gate-length scaling; found SOI MOSFETs exhibit significantly lower leakage current and reduced DIBL versus aggressively scaled bulk devices, validated against measured SOI data.
- **PN Junction Simulation & Numerical Analysis (MATLAB)** - Modelled PN junction electrostatics (electric field, potential, depletion width) across doping concentration ratios; validated numerical results against analytical theory, confirming convergence was independent of mesh discretization scheme.
- **Heat Flow Simulation in SOI Wafers (COMSOL Multiphysics)** - Built 2D/3D thermal models of localized transistor heating in SOI structures; found peak temperatures of **$\sim 336\text{ K}$** beneath the heat source and established thermal convergence for wafer widths above **42.5 μm** .

PUBLICATIONS

Comparative Approach for the Design of Delay Lines TDC Architecture. NCCS, Springer Journal, June 2020. Co-authored research achieving 9.76 ps timing precision using Xilinx ISE.

AWARDS & LEADERSHIP

Distinction - CeNSE Winter School, IISc Bangalore (2025)

Awarded Distinction in the examination at India's premier nanoscience research institute, demonstrating exceptional depth in nanofabrication and device physics.

Semi-Finalist - Texas Instruments India Innovation Challenge (2019)

Prototyped SMARTKART, an IoT-integrated automated checkout system (TIVA & MSP microcontroller tracks) - recognised among top student innovators nationally.

International Student Ambassador, KTH

Selected to guide incoming international students through programme selection and onboarding, demonstrating leadership and cross-cultural communication skills.